

Date: 06/08/2026

Object Oriented Java Programming Lab Exercises

Exercise 1: Java Program Fundamentals

AIM:

To write a Java Program using **variables, data types, operators, input/output statements, and control structures.**

ALGORITHM:

- 1) Start the program.
- 2) Create a **Scanner** object to read input from the user.
- 3) Declare variables of different data types (**int, double, char, boolean**).
- 4) Read two integer values from the user.
- 5) Perform arithmetic operations:
 - Addition
 - Subtraction
 - Multiplication
 - Division (If the second number is not **Zero**).
 - Modulus
- 6) Calculate the average of the two numbers.
- 7) Compare the two numbers using a relational operator.
- 8) Determine the grade using the **if-else** control structure based on the average.
- 9) Display the results of all operations.
- 10) Use a **for** loop to display numbers from 1 to 10.
- 11) Close the **Scanner** object.
- 12) Stop the program.

SOURCE CODE:

```
import java.util.Scanner;
public class Exercise1 {
    public static void main(String[] args) {
        //Scanner Object
        Scanner sc = new Scanner(System.in);

        //Variable Declaration
        int num1, num2;
        double average;
        char grade;
        boolean isGreater;

        //Inputs
        System.out.print("Enter first number: ");
        num1 = sc.nextInt();
        System.out.print("Enter second number: ");
        num2 = sc.nextInt();

        //Operations
        int sum = num1 + num2;
        int difference = num1 - num2;
        int product = num1 * num2;

        double quotient;

        if (num2 != 0) {
            quotient = (double) num1 / num2;
        } else {
            System.out.println("Division by zero is not
allowed.");
            quotient = 0; // Assigning a default value
        }

        average = (num1 + num2) / 2.0;

        //Relational Operation
        isGreater = num1 > num2;

        //Grade Calculation - IF ELSE
        if (average >= 90) {
            grade = 'A';
        } else if (average >= 80) {
            grade = 'B';
        } else if (average >= 70) {
            grade = 'C';
        } else if (average >= 60) {
            grade = 'D';
        } else {
            grade = 'F';
        }
    }
}
```

```

//Outputs
System.out.println("Sum: " + sum);
System.out.println("Difference: " + difference);
System.out.println("Product: " + product);
if (num2 != 0) {
    System.out.println("Quotient: " + quotient);
} else {
    System.out.println("Quotient: Undefined (division
                        by zero)");
}
System.out.println("Average: " + average);
System.out.println("Is first number greater than
                    second? " + isGreater);
System.out.println("Grade: " + grade);

//Loop - For Loop
System.out.println("Numbers from 1 to 10:");
for (int i = 1; i <= 10; i++) {
    System.out.print(i + " ");
}
System.out.println(); // Print a new line after the
                        numbers

//Closing Scanner
sc.close();
    }
}

```

COMPILATION PROCESS:

1. Type the java Program in a programming environment and save it with the file name **Exercise1.java**.
2. Open the Command Prompt or Terminal.
3. Navigate to the folder where the program is saved using the **cd** command.
4. Compile the program using the command: `javac Exercise1.java`
5. If no error occurred, then Run the program using the command: `java Exercise1`

OUTPUT:

```
• shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ javac Program1.java
• shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ java Program1
Enter first number: 18
Enter second number: 27
Sum: 45
Difference: -9
Product: 486
Quotient: 0.6666666666666666
Average: 22.5
Is first number greater than second? false
Grade: F
Numbers from 1 to 10:
1 2 3 4 5 6 7 8 9 10
```

RESULT:

The Java Program was **compiled and executed successfully**. The Expected output has been generated and verified.

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Exercise 2: Arrays and Strings

AIM:

To develop a Java Program using **one-dimensional array, two-dimensional array, String, and StringBuffer** classes.

ALGORITHM:

1. Start the program.
2. Create a **Scanner** object to read input from the user.
3. Declare a one-dimensional array of size 5.
4. Read five integer elements into the array.
5. Calculate the sum of the array elements.
6. Display the elements and their sum.
7. Declare a two-dimensional (2x2) array.
8. Read the elements of the 2x2 matrix from the user.
9. Display the matrix.
10. Read a string from the user.
11. Perform string operations:
 - Find the length of the string.
 - Convert it to uppercase.
 - Convert it to lowercase.
 - Reverse the string.
12. Create a **StringBuffer** object and Perform the following operations:
 - Append a string.
 - Insert a string.
 - Replace characters.

- Delete characters.
- Reverse the string.

13. Display the results of all operations.

14. Close the **Scanner** object.

15. Stop the program.

SOURCE CODE:

```
import java.util.Scanner;
public class Exercise2 {
    public static void main(String[] args) {
        //Scanner Object
        Scanner sc = new Scanner(System.in);
        //----- 1D Array -----
        int[] arr = new int[5];
        int sum = 0;
        System.out.println("Enter 5 numbers for the array:");
        for (int i = 0; i < arr.length; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        System.out.println("\nElements in the array:");
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }

        System.out.println("\nSum of array elements: " + sum);

        //----- 2D Array -----
        int[][] matrix = new int[2][2];
        System.out.println("\nEnter elements for a 2x2 matrix:");
        for (int i = 0; i < matrix.length; i++) {
            for (int j = 0; j < matrix[i].length; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        System.out.println("\nElements in the 2x2 matrix:");
        for (int i = 0; i < matrix.length; i++) {
            for (int j = 0; j < matrix[i].length; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }

        //----- String Operations -----
        System.out.print("\nEnter a string: ");
```

```

String inputString = sc.next();
System.out.println("You entered: " + inputString);
System.out.println("Length of the string: " +
                    inputString.length());
System.out.println("Uppercase: " +
                    inputString.toUpperCase());
System.out.println("Lowercase: " +
                    inputString.toLowerCase());
System.out.println("Reversed string: " + new
                    StringBuilder(inputString).reverse());

//----- StringBuffer Operations -----
StringBuffer sb = new StringBuffer(inputString);
sb.append(" Java");
System.out.println("\nStringBuffer after appending: " +
                    sb);

sb.insert(0, "Hello ");
System.out.println("StringBuffer after inserting: " +
                    sb);

sb.delete(0, 6);
System.out.println("StringBuffer after deleting: " +
                    sb);

sb.replace(0, 4, "Hi");
System.out.println("StringBuffer after replacing: " +
                    sb);

sb.reverse();
System.out.println("StringBuffer after reversing: " +
                    sb);

sc.close();
    }
}

```

COMPILATION PROCESS:

1. Type the java Program in a programming environment and save it with the file name **Exercise2.java**.
2. Open the Command Prompt or Terminal.
3. Navigate to the folder where the program is saved using the **cd** command.
4. Compile the program using the command: `javac Exercise2.java`
5. If no error occurred, then Run the program using the command: `java Exercise2`

OUTPUT:

```
1 2 3 4 5 6 7 8 9 10
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ javac Program2.java
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ java Program2
Enter 5 numbers for the array:
1 4 2 5 6

Elements in the array:
1 4 2 5 6
Sum of array elements: 18

Enter elements for a 2x2 matrix:
1 5 2 8

Elements in the 2x2 matrix:
1 5
2 8

Enter a string: Hello
You entered: Hello
Length of the string: 5
Uppercase: HELLO
Lowercase: hello
Reversed string: olleH

StringBuffer after appending: Hello Java
StringBuffer after inserting: Hello Hello Java
StringBuffer after deleting: Hello Java
StringBuffer after replacing: Hio Java
StringBuffer after reversing: avaJ oiH
```

RESULT:

The Java Program was **compiled and executed successfully**. The Expected output has been generated and verified.

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Exercise 3: Classes and Object

AIM:

To develop a Java Program demonstrating **Classes, Objects, Methods, and Constructors**

ALGORITHM:

1. Start the program.
2. Import the **Scanner** class for the user input.
3. Define a **Student** class with data members:
 - Roll Number
 - Name
 - Marks array
 - Total and Average
 - Grade
4. Create a parameterized constructor to initialize the student's roll number and name.
5. Create the **getMarks()** method to
 - Read marks for 5 subjects.
 - Store them in an array.
 - Calculate the total marks.
 - Calculate the average marks.
6. Create the **calculateGrade()** method to determine the grade based on the average.
7. Create the **displayDetails()** method to display the student's roll number, name, subject marks, total, average, and grade.
8. In the **main()** method:

- Read the student's roll number and name.
 - Create a **student** object using the constructor.
 - Call the **getMarks()** method.
 - Call the **calculateGrade()** method.
 - Call the **displayDetails()** method.
9. Close the **Scanner** object.
 10. Stop the program.

SOURCE CODE:

```
import java.util.Scanner;

class Student {
    int rollNo;
    String name;
    int[] marks = new int[5];
    int total;
    double average;
    char grade;

    Student(int rollNo, String name){
        this.rollNo = rollNo;
        this.name = name;
    }

    //methods to get marks
    void getMarks(Scanner sc){
        total = 0;
        System.out.println("\nEnter marks for 5 subjects: ");
        for(int i=0; i<5; i++){
            System.out.println("Subject " + (i+1) + ": ");
            marks[i] = sc.nextInt();
            total += marks[i];
        }
        average = (double)total / marks.length;
    }

    //calculate grade
    void calculateGrade(){
        if(average >= 90){
            grade = 'A';
        } else if(average >= 80){
            grade = 'B';
        } else if(average >= 70){
            grade = 'C';
        } else if(average >= 60){
```

```

        grade = 'D';
    } else {
        grade = 'F';
    }
}

//display student details
void displayDetails(){
    System.out.println("\nStudent Details:");
    System.out.println("Roll No: " + rollNo);
    System.out.println("Name: " + name);
    System.out.println("Marks: ");
    for(int i=0; i<marks.length; i++){
        System.out.println("Subject " + (i+1) + ": " +
marks[i]);
    }
    System.out.println("Total Marks: " + total);
    System.out.println("Average Marks: " + average);
    System.out.println("Grade: " + grade);
}

}

public class Exercise3 {
    public static void main(String[] args) {
        //Scanner Object Creation
        Scanner sc = new Scanner(System.in);

        //User Inputs
        System.out.print("Enter roll number: ");
        int rollNo = sc.nextInt();

        sc.nextLine(); // consume the newline character

        System.out.print("Enter name: ");
        String name = sc.nextLine();

        //Object Creation using parameterized Constructor
        Student student = new Student(rollNo, name);

        //Class Methods
        student.getMarks(sc);
        student.calculateGrade();
        student.displayDetails();

        //Scanner Object Closed
        sc.close();
    }
}

```

COMPILATION STEPS:

1. Type the java Program in a programming environment and save it with the file name **Exercise3.java**.
2. Open the Command Prompt or Terminal.
3. Navigate to the folder where the program is saved using the **cd** command.
4. Compile the program using the command: `javac Exercise3.java`
5. If no error occurred, then Run the program using the command: `java Exercise3`

OUTPUT:

```
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ javac Program3.java
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ java Program3
Enter roll number: 101
Enter name: Ram

Enter marks for 5 subjects:
Subject 1:
80
Subject 2:
83
Subject 3:
72
Subject 4:
90
Subject 5:
92

Student Details:
Roll No: 101
Name: Ram
Marks:
Subject 1: 80
Subject 2: 83
Subject 3: 72
Subject 4: 90
Subject 5: 92
Total Marks: 417
Average Marks: 83.4
Grade: B
```

RESULT:

The Java Program was **compiled and executed successfully**. The Expected output has been generated and verified.

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Exercise 4: Inheritance and Polymorphism

AIM:

To develop a Java Program demonstrating the concepts of **Single Inheritance**, **Multilevel Inheritance**, **Hierarchical Inheritance**, **Method Overloading**, and **Method Overriding**.

ALGORITHM:

1. Start the program.
2. Create a superclass named **Person** with a data member **name** and a method **display()**.
3. Create a subclass **Student** that inherits from **Person**.
4. Override the **display()** method in the **Student** class.
5. Create two overloaded **showDetails()** methods in the **Student** class.
6. Create another subclass **SportsStudent** that inherits from **Student** to demonstrate Multilevel Inheritance.
7. Override the **display()** method in the **SportsStudent** class.
8. Create another subclass **Teacher** that inherits from **Person** to demonstrate Hierarchical Inheritance.
9. Override the **display()** method in the **Teacher** class.
10. In the **main()** method, create objects of the **Student**, **SportsStudent**, and **Teacher** classes.
11. Call the appropriate methods to demonstrate Inheritance, Method Overloading, and Method Overriding.
12. Display the output.
13. Stop the Program.

SOURCE CODE:

```
class Person{
    String name;

    Person(String name) {
        this.name = name;
    }

    void display() {
        System.out.println("Name: " + name);
    }
}

//Single Inheritance
class Student extends Person {
    int rollNo;

    Student(String name, int rollNo) {
        super(name);
        this.rollNo = rollNo;
    }

    //Method Overriding
    @Override
    void display() {
        super.display();
        System.out.println("Roll No: " + rollNo);
    }

    //Method Overloading
    void showDetails(){
        System.out.println("Student details displayed
successfully.");
    }

    void showDetails(String dept){
        System.out.println("Department: " + dept);
    }
}

//Multilevel Inheritance
class SportsStudent extends Student {
    String sport;

    SportsStudent(String name, int rollNo, String sport) {
        super(name, rollNo);
        this.sport = sport;
    }

    @Override
    void display() {
        super.display();
    }
}
```

```

        System.out.println("Sport: " + sport);
    }
}

//Hierarchical Inheritance
class Teacher extends Person {
    String subject;
    Teacher(String name, String subject) {
        super(name);
        this.subject = subject;
    }

    @Override
    void display() {
        super.display();
        System.out.println("Subject: " + subject);
    }
}

public class Program4 {
    public static void main(String[] args) {
        System.out.println("==== Student =====");
        Student student = new Student("John Doe", 101);
        student.display();
        student.showDetails();
        student.showDetails("Computer Science");

        System.out.println("\n==== Sports Student =====");
        SportsStudent sportsStudent = new SportsStudent("Jane
Smith",
"Basketball");
        sportsStudent.display();

        System.out.println("\n==== Teacher =====");
        Teacher teacher = new Teacher("Mr. Johnson",
"Mathematics");
        teacher.display();
    }
}

```

COMPILATION STEPS:

1. Type the java Program in a programming environment and save it with the file name **Exercise4.java**.
2. Open the Command Prompt or Terminal.
3. Navigate to the folder where the program is saved using the **cd** command.
4. Compile the program using the command: `javac Exercise4.java`

5. If no error occurred, then Run the program using the command: java Exercise4

OUTPUT:

```
Grade: D
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ javac Program4.java
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs$ java Program4
===== Student =====
Name: John Doe
Roll No: 101
Student details displayed successfully.
Department: Computer Science

===== Sports Student =====
Name: Jane Smith
Roll No: 102
Sport: Basketball

===== Teacher =====
Name: Mr. Johnson
Subject: Mathematics
```

RESULT:

The Java Program was **compiled and executed successfully**. The Expected output has been generated and verified.

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Exercise 5: Packages and Interfaces

AIM:

To develop a Java Program using an **Interface** and a **User-defined Package** to perform simple Arithmetic operations.

ALGORITHM:

1. Start the program.
2. Create an interface **Calculator** inside a package named **mypackage**.
3. Declare methods in the interface:
 - **add()**
 - **subtract()**
 - **product()**
 - **divide()**
 - **modulus()**
4. Import the **Calculator** interface into the main program.
5. Create a class **MyCalculator** that implements the **Calculator** interface.
6. Override the methods inside the class **MyCalculator**.
7. In the main method:
 - Declare a **Scanner** object and read two integer values from user.
 - Create an object of the class **MyCalculator**
 - Call the **Overridden** methods using the class object to perform the operations.
8. Display the results.
9. Stop the program.

STRUCTURE:

```
Program_Folder/  
|  
|- Exercise4.java  
|- MyPackage/  
|  
|- Calculator.java
```

SOURCE CODE:

Program_Folder/MyPackage/Calculator.java:

```
package MyPackage;  
  
public interface Calculator {  
    public int add(int a, int b);  
    public int subtract(int a, int b);  
    public int multiply(int a, int b);  
    public double divide(int a, int b);  
    public int modulus(int a, int b);  
}
```

Program_Folder/Exercise5.java:

```
import MyPackage.Calculator;  
import java.util.Scanner;  
class MyCalculator implements Calculator {  
    @Override  
    public int add(int a, int b) {  
        return a + b;  
    }  
  
    @Override  
    public int subtract(int a, int b) {  
        return a - b;  
    }  
  
    @Override  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
  
    @Override  
    public double divide(int a, int b) {  
        if (b == 0) {  
            System.out.println("Error: Division by zero is  
                                not allowed.");  
        }  
        return (double) a / b;  
    }  
}
```

```

        @Override
        public int modulus(int a, int b) {
            if (b == 0) {
                System.out.println("Error: Modulus by zero is not
                                   allowed.");
                return 0;
            } else {
                return a % b;
            }
        }
    }

    public class Exercise5 {
        public static void main(String[] args) {
            Scanner sc = new Scanner(System.in);

            System.out.print("Enter first number: ");
            int num1 = sc.nextInt();

            System.out.print("Enter second number: ");
            int num2 = sc.nextInt();

            MyCalculator calculator = new MyCalculator();

            System.out.println("Addition: " + calculator.add(num1,
                                                                num2));
            System.out.println("Subtraction: " +
                               calculator.subtract(num1, num2));
            System.out.println("Multiplication: " +
                               calculator.multiply(num1, num2));

            if(num2 != 0) {
                System.out.println("Division: " +
                                    calculator.divide(num1, num2));
            } else {
                System.out.println("Division: Error: Division by
                                    zero is not allowed.");
            }

            if(num2 != 0) {
                System.out.println("Modulus: " +
                                    calculator.modulus(num1, num2));
            } else {
                System.out.println("Modulus: Error: Modulus by
                                    zero is not allowed.");
            }

            sc.close();
        }
    }
}

```

COMPILATION STEPS:

1. Type the java Program in a programming environment and save it with the file name **Exercise5.java**.
2. Open the Command Prompt or Terminal.
3. First change the directory to the Package **MyPackage** using the **cd**, and compile the **Calculator.java** by using the command: **javac Calculator.java**.
4. Then Change back to the parent directory, and compile the file **Exercise5.java** using the command: **javac Exercise5.java**
5. Execute the code Exercise5.java using the command: **java Exercise5**.

OUTPUT:

```
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs/Program5$ cd MyPackage
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs/Program5/MyPackage$ javac Calculator.java
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs/Program5/MyPackage$ cd ..
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs/Program5$ javac Program5.java
shadow2202@shadow2202-Latitude-5420:~/CyberLabs/Java/Lab_Programs/Program5$ java Program5
Enter first number: 28
Enter second number: 15
Addition: 43
Subtraction: 13
Multiplication: 420
Division: 1.8666666666666667
Modulus: 13
```

RESULT:

The Java Program was **compiled and executed successfully**. The Expected output has been generated and verified.